

Corporate university model: enhancing competitiveness in firms from emerging economies

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Abstract

Purpose – Corporate universities (CUs) serve as a strategy to strengthen corporate competitive advantage in developed economies. Their effectiveness is related to knowledge management (KM), the adoption of emerging technologies and collaborative work (CW). However, in current models of CUs, these aspects are partially and individually considered, and no model focuses on firms in emerging economies. Therefore, this study aims to address this gap by setting forth a holistic model of CUs by considering KM, technologies, such as Industry 4.0 (I4.0) and Industry 5.0 (I5.0) technologies, and CW concurrently to enhance firms' sustainable competitiveness in Latin America.

Design/methodology/approach – The model proposed in this paper was developed after a comprehensive systematic literature review and subsequent content analysis was conducted on 64 articles sourced from the Web of Science and Scopus databases, exploring publications from 1995 to 2025.

Findings – The interrelation among KM, technologies such as I4.0 and I5.0 and CW supports the CU model, as it enables continuous innovation, efficient KM processes and adaptability to dynamic market conditions.

Research limitations/implications – CU is a constant changing phenomenon, it is suggested to validate the proposed model and consider aspects such as corporate university (CU) types and KM environments in emerging economies and to compare it with developed economies.

Practical implications – The model could provide firms a practical tool to help design, implement and manage CU with an emphasis on KM, Industry 4.0 and Industry 5.0 technologies and CW.

Social implications – The forms of KM in CU are changing over time. Human talent is essential in KM. It is crucial to take into account the factors influencing training and development of their abilities.

Originality/value – This study contributes to KM and I4.0/I5.0. It demonstrates how the CU model fosters firms' competitive advantage in emerging economies while enabling knowledge creation and exchange with stakeholders and society. By integrating KM activities and I4.0/I5.0, this study advances the development of a CU model for Latin American and Caribbean firms, strengthening their human resource capabilities.

Keywords Corporate universities, Knowledge management, Industry 4.0 and Industry 5.0 technologies, Collaborative work, Sustainable competitive advantages, Emerging economies

Paper type Conceptual paper

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1. Introduction

Today, CU represents an innovative strategy applied in organizations in developed economies to address the challenges imposed by the constant and complex environmental changes (Chen et al., 2019; Kolo et al., 2013). The literature confirms that CUs are important for firms, as they support the development of talent and the strategy for achieving their objectives (Araujo & Domingos, 2020; Lissillour et al., 2020). These models have also been found to strengthen the sustainable competitive advantages of organizations (Alonso-Gonzalez et al., 2018; Han et al., 2024); develop human talent; promote the use of technology (Chen & Xu, 2022; Lui Abel & Li, 2012); foster knowledge management (KM) and organizational learning. In addition, these models have been demonstrated to develop technological, social and cultural processes (Khan, 2021; Prince & Stewart, 2002); increase productivity through attraction, training and retention of employees and customers (Ilyas, 2017); and generate learning networks with their stakeholders, to strengthen employees (Singh et al., 2020).

The literature includes studies on CUs in China (Chen et al., 2019, 2022; Chen & Xu, 2022; Han et al., 2024; Lissillour & Rodriguez-Escobar, 2023), Indonesia (Ayuningtias et al., 2016), Japan (Nguyen & Fan, 2022), Italy (Cappiello & Pedrini, 2017), the USA (Lui Abel & Li, 2012; Vives, 2015) and numerous firms with participation in global markets (Margherita & Secundo, 2011). This research conducted a systematic literature review (SLR) and identified 19 CU models (Table A1, Appendix, and Supplementary Table 3), 17 of which focused on developed economies. On the other hand, although some studies demonstrate how CUs can enhance personnel development in emerging economies (Han et al., 2024), only three models provide empirical evidence in this context: two in Brazil (Araujo & Domingos, 2020) and one in Colombia (Alonso-Gonzalez, Peris-Ortiz, & Palacios Chacon, 2018). This gap is particularly relevant considering that firms in emerging economies often operate under labor-intensive conditions, with low employee qualifications, limited investment in KM, scarce adoption of emerging technologies and persistently low labor productivity (Bernal-Torres et al., 2023; Gölgeci et al., 2019).

This SLR highlights that KM, emerging technologies and collaborative work (CW) are the basis for the effectiveness of CU models (Alonso-Gonzalez et al., 2018; Nguyen & Fan, 2022). In addition, this study demonstrates that from the 19 CU models, in 16 the KM position KM as a critical factor in their activities (see Supplementary Table 3 for KM activities). However, in four of these models, attention is limited to the knowledge transfer activity (Bresolin & De Sá Freire, 2023; Chen et al., 2022); three of these models accentuate knowledge creation and exchange (Chen et al., 2023; Prince & Stewart, 2002), and in two models, knowledge generation, exchange, optimization and transformation are emphasized (Han et al., 2024). In addition, it is notable that KM is partly performed in the existing CU models without considering all KM activities simultaneously (identification, recording, socialization, transfer, creation, use, incubation and exploitation of knowledge). Thus, these models are limited in that they do not provide a comprehensive view of KM (identification, recording, socialization, transfer, creation, use, incubation and exploitation of knowledge) in CUs (Chen et al., 2019, 2022).

On the other hand, in terms of technology, 15 models considered information and communication technologies (ICTs) as a success factor, which evidences the relevance of these technologies in the dynamics of CUs. However, new technologies, which predominantly refer to Industry 4.0 (I4.0) and Industry 5.0 (I5.0) have not been integrated in the CU models, overlooking the ways in which these technologies contribute to organizational competitive advantage and improved employee performance (Lista Rossetti et al., 2024). Therefore, a CU model is required to emphasize the use and appropriation of ICTs as a company strategy to create a competitive advantage.

Finally, 12 models emphasize intra- and inter-organizational CW as a significant driver of CU activities. Seven models highlight the importance of both inter- and intra-organizational collaboration for achieving corporate outcomes (Margherita & Secundo, 2011). In this vein, CW has been demonstrated to facilitate the creation and transfer of knowledge, social innovation, it encourages intrapreneurship; and supports strategic objectives. Without such collaboration, firms face fewer opportunities to respond effectively to market dynamics (Chen et al., 2019). However, no existing model promotes a comprehensive and collaborative vision for CUs that integrates different actors (firms, academic institutions, government entities and other stakeholders). Therefore, by incorporating these actors into a CU model, organizations will be provided with a blueprint for enhanced resource optimization and sustainable competitive advantages (Han et al., 2024).

Overall, this SLR demonstrates that KM, ICTs and CW are key factors in CUs to enhance the competitive capacity of organizations, but existing models consider these factors in a partial way, which limits their effectiveness, a situation much more evident in CUs in emerging economies. To address this gap in the literature, this study proposes an integrated CU model for firms in emerging economies of Latin America, supported by KM, covering all activities comprehensively, I4.0 and I5.0 (including ICT) and CW (broadly).

The rest of the paper is organized as follows. Section 2 describes the study's methodology and SLR findings are presented in Section 3. Section 4 examines these findings and introduces a holistic CU model based on the literature review. The discussion is presented in Section 5. Finally, the paper discusses the results and concludes in Section 6.

2. Methodology

The proposed CU model was designed in two stages: first, an SLR on CUs was conducted to develop the literature review (Section 3) as the basis of the CU model. Second, the model proposal was developed (Section 4).

2.1 Stage 1: systematic literature review on corporate university, corporate university models and relevant factors

Between February 2023 and February 2024, searches were conducted on Web of Science and Scopus to identify literature on CUs and CU models. This was done in line with the preferred reporting items for systematic reviews and meta-analyses (PRISMA) protocol proposed by Tranfield, Denyer, and Smart (2003) and Page et al. (2021) (Figure 1):

- Step 1: Search strategy.

The researchers used Cerchione and Esposito (2016) guidelines to select databases, keywords and construct the search string. The researchers chose Web of Science and Scopus databases (DBs), owing to their comprehensive coverage, robustness, classification and refinement features. The review started with the terms “CUs” and “CU models”, later, “KM” was added after identifying the relationship between “CUs”, “CU models” and “KM”. This led to the use of four search equations:

E1: “corporate universit*”.

E2: “corporate universit* model” or “corporate university* model”.

E3: “corporate universit*” and “knowledge management”.

E4: “corporate universit*” and “knowledge management” and “model” or “framework”.

The search parameters considered articles, book chapters and conference proceedings, resulting in 4,246 records. After the removal of duplicates and excluding records without

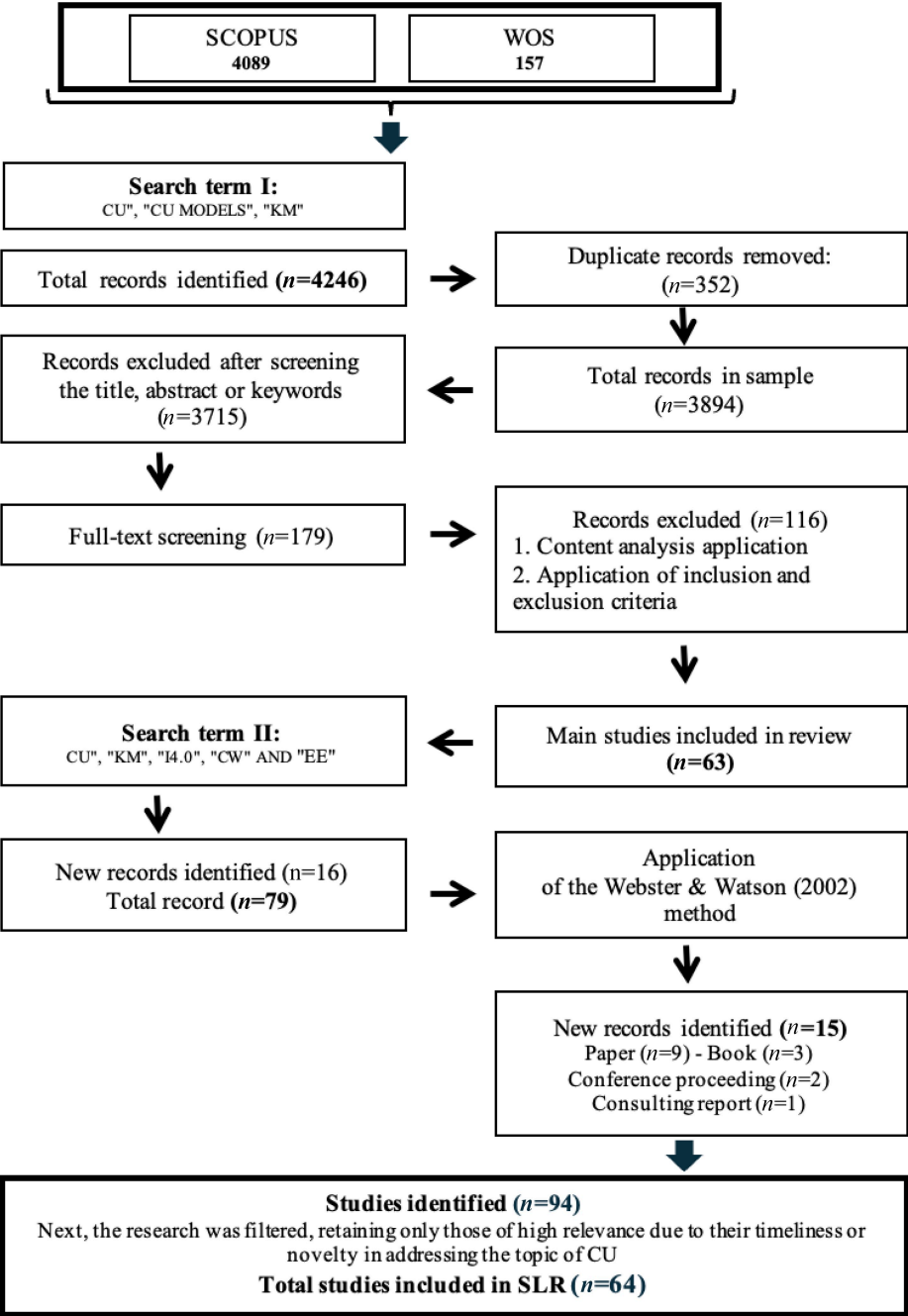


Figure 1. PRISMA flow diagram summarizing source identification and screening steps
Source: Authors' own work

“CUs”, “KM” and “CU model” in the title, abstract or keywords, 179 records formed part of the sample.

- Step 2: Research gaps and objectives.

[Sandberg and Alvesson's \(2011\)](#) gap-spotting logic was applied to explore CUs, KM and CU models in the 179 articles.

- Step 3: Inclusion and exclusion criteria and data analysis.

To ensure that the articles in the SLR were relevant, the following inclusion and exclusion criteria were applied (Supplementary Table 1):

- The “content analysis” method ([Mayring, 2015](#)) was used, following descriptive analysis, category selection and material evaluation. Additionally, qualitative descriptive coding was applied according to [Cerchione and Esposito \(2016\)](#) guidelines, with abstracts assisting in consolidating research ideas and opinions.
- Keywords and term relations were identified, which helped discern research gaps ([Liao et al., 2018](#)). Likewise, keyword co-occurrence analysis (author/index) and content analysis of text data (title and abstracts) were performed using the VOSviewer (Supplementary Figures 1 and 2).
- The field base/evolution was studied, and reference co-citation analysis ([Walsh & Renaud, 2017](#)) was adopted using VOSviewer (Supplementary Figure 3).

After applying the inclusion and exclusion criteria, the sample comprised 63 records.

- Step 4: New DB search.

As a result of the content analysis, four relevant factors (KM, emerging technologies -including I4.0 and I5.0, CW, and emerging economies) were identified in the context of CUs. This led to a new search for records in the DB, using the following search equations:

E5: “corporate universities” or “corporate university” and “I4.0” and “I5.0”.

E6: “corporate universities” or “corporate university” and “collaborative work” or “collaboration”.

E7: “knowledge management” and “corporate universities” or “corporate university” and “emerging markets” or “emerging economies”.

This process generated 16 new records addressing: CUs, KM, I4.0, I5.0 CW and EE, 79 records remained.

- Step 5: Inclusion of seminal studies.

To ensure the inclusion of relevant research, the “forward-backward” method ([Watson & Webster, 2020](#)) was implemented, which resulted in the addition of 15 records to the sample, including two non-English studies. Finally, after a thorough analysis of the 94 studies identified, the research was filtered, retaining only those of high relevance because of their timeliness or novelty in addressing the topic of CUs, or because the topics analyzed on CU were presented in various studies. This resulted in a final sample of 64 studies (Supplementary Table 2), including the corresponding references and 20 additional studies from the reviewers’ amendments and those used in the methodology.

2.2 Stage 2: systematic literature review results

The VOS viewer software was used to analyze co-occurrence and text data from titles and abstracts (Supplementary Figures 1, 2 and 3), identifying CU-related factors. Additionally,

3. Literature review

Source: Authors' own work

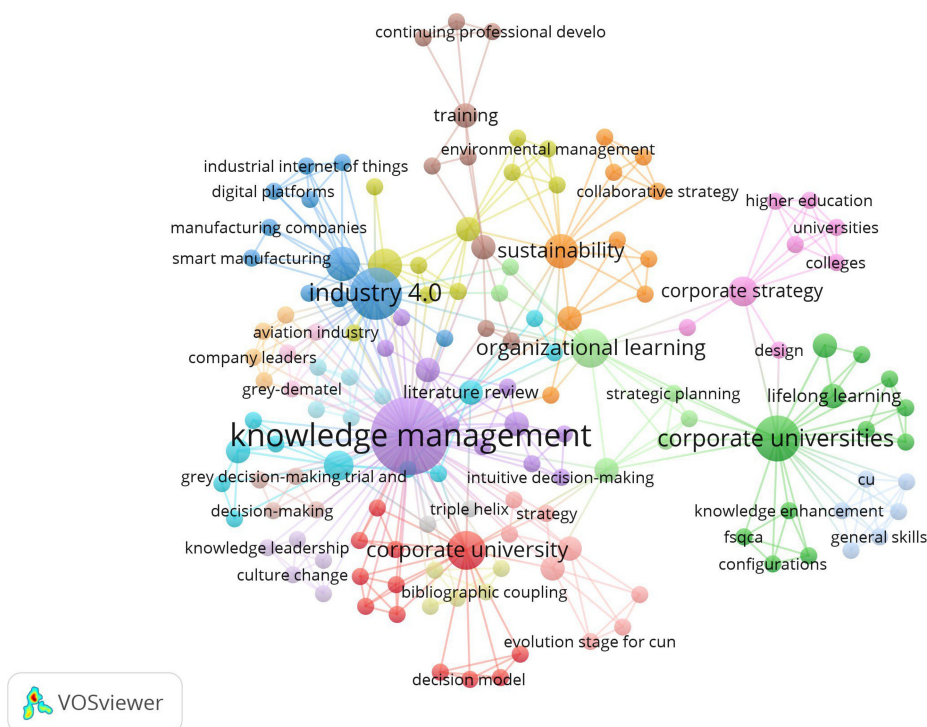


Figure 3. Map based on co-occurrence – authors keywords
Source: Authors' own work

3.1 Corporate universities

CUs originated in the USA in 1955 and gained popularity in the 1980s (Meister, 1998). Early adopters included General Motors, General Electric and McDonald's. Today, global corporations such as Apple, Boeing and Danone have CUs (McKinsey, 2016). By 2019, China comprised around 3,700 CUs, and 80% of the firms listed in the Fortune Global 500 had established CUs (Chen et al., 2019; Lissillour et al., 2020).

Initially, CUs were conceived as strategic institutions for staff training (Meister, 1998) and learning activities (Allen, 2010). Today, CUs serve as an organizational strategy (Aziz & George, 2020) implemented to enhance sustainable competitive advantage through KM practices (Chen et al., 2023; Chen & Xu, 2022; Scarso, 2017). In addition, CUs function as educational units for achieving corporate objectives by developing intellectual capital (Singh et al., 2020), they also facilitate the transfer and accumulation of internal and external knowledge (Chin et al., 2019), and they serve as a basis for training and learning initiatives (Ayuningtias et al., 2016).

According to the literature, CUs are promoted by human resources departments, executives, or learning and development (L&D) departments (Asalla et al., 2019). They facilitate collective and individual learning through networks and knowledge integration (Lissillour & Rodriguez-Escobar, 2023). CUs have been demonstrated to enhance competitiveness and decision-making (Aziz & George, 2020), and they aid in aligning

organizational goals with personnel management (Lui Abel & Li, 2012). In addition, CUs have been found to promote innovation and strategic alliances (Allen, 2010; Dealtry & Howard, 2008), skill development by way of technology (El-Tannir, 2002) and ensure competitive advantage via knowledge transfer (Meister, 1998).

Considering the above and given the lack of consensus in the literature regarding the definition and functions of CU, this study proposes a definition of CU. This definition has been cultivated by triangulating the definitions of CUs identified in the SLR (Supplementary Table 3), including those presented by Meister (1998), as well as those described by Chen, Zhou, and Wang (2023) and Han et al. (2024). Additionally, a thematic analysis of these definitions was conducted, integrating them into a coherent and updated proposal. Thus, the present research defines as CU as: *a corporate strategy aimed at enhancing the firm's sustainable competitive advantage, adding value for its stakeholders through effective KM, and the appropriate use and integration of emerging technologies, all within CW between the firm and its stakeholders.*

The literature emphasizes the significant role of CUs in firms and demonstrates that CUs address learning needs, they support knowledge dynamics (Han et al., 2024), ensure the supply of highly skilled personnel (Aziz & George, 2020), enhance social innovation through stakeholder collaboration and increase employee engagement (Chin et al., 2019). Thus, successful CUs align organizational strategies with dynamic, competitive environments (Chen & Xu, 2022; Lissillour et al., 2020; Lui Abel & Li, 2012), which has resulted in increasing interest from consulting firms. For example, Deloitte (2022) reported that 80% of firms succeed when investing in employee training, as it has been shown to boost innovation, reduce employee turnover and enhance organizational reputation. McKinsey (2016) highlighted that CUs foster collaborative learning and networking through digital transformation, enabling swift responses to market dynamics. Similarly, Boston Consulting Group (2020) demonstrated that firms should adopt comprehensive corporate learning systems via CUs to develop a competitive advantage focused on learning capabilities.

Han, Zhou, Carr, and Jiang (2024) state that CUs should empower firms, guiding their KM and cultural transformation processes. This enables firms to respond to and shape external demands, benefiting stakeholders. Additionally, the literature emphasizes that firms must address workforce competency gaps, especially in response to disruptive technological advances (Saniuk et al., 2024).

3.2 Corporate university models

Supplementary Table 3, in the supplementary information section of this paper, outlines 19 CU models identified in the SLR, detailing their key characteristics, components and contributions to firms. This Table highlights the organizational theories underpinning these models, predominantly KM and organizational learning. Of the 19 models, 13 focused on KM activities and five of them focused on knowledge creation and transfer through stakeholder relationships, distinguishing between tacit and explicit knowledge and linking formal knowledge with practice. In addition, these models were found to promote learning, innovation and improved employee and organizational performance (Asher & Popper, 2021; Nonaka et al., 2000). Two models emphasized organizational learning, particularly lifelong learning and skill development (Chen et al., 2023; Han et al., 2024). Only 12 models were validated through case studies and grounded theory. Table A1 in Appendix shows the factors (KM, technology, I4.0 and CW) in the 19 CU models. This resulted in a final sample of 19 CU models (Supplementary Table 2).

3.3 *Corporate university and knowledge management relation*

Research shows that firms increasingly seek CU to integrate KM as a core function and take on broader roles (Chen, Xu, & Zhai, 2019). CUs represent the ideal strategy and structure for establishing effective KM systems (Chen et al., 2023; Scarso, 2017) and enhancing various activities (Asalla, Hamsal, & Budiastuti, 2019). KM has been found to promote knowledge exchange, innovation and implementation (Lissillour et al., 2020; Rhéaume & Gardoni, 2016); facilitate industry-university-research cooperation (Scarso, 2017); and optimize governance and knowledge-sharing strategies (Han et al., 2024). In China, CUs were found to enhance innovation via government-firm cooperation (Chen & Xu, 2022; Chin et al., 2019). Research on CUs also demonstrates how KM promotes the identification, socialization, transfer and exploitation of knowledge. However, no study fully incorporates all KM activities (Chen et al., 2019; Scarso, 2017). Additionally, no research addresses KM holistically, linking internal and external knowledge sources (Scarso, 2017), which underscores the need for further investigation (Chen & Xu, 2022).

3.4 *Corporate university and emerging technologies*

Literature on CU models presents the relationship between technologies I4.0 and KM practices, highlighting the need for firms to develop specific KM skills at individual, team and organizational levels (Tortorella et al., 2020). I4.0, such as Internet of Things (IoT), big data, has been found to contribute to processes of knowledge creation, organization and implementation (Lista Rossetti, Luz Tortorella, Bouzon, Gao, & Chan, 2024). These technologies facilitate KM (capture, creation, transfer and reuse) (Asalla et al., 2023) and help firms transform information into knowledge and promote sustainable competitive advantages (Chen et al., 2023). Digital transformation introduces advanced technologies such as I5.0, which builds on the foundations of I4.0 (Pasumarthy et al., 2024). A key aspect of I5.0 is its emphasis on human resources, integrating technology with workforce management and reinforcing organizational efficiency (Saniuk et al., 2024). This demonstrates that further studies should guide the development of I4.0 capabilities in CUs.

3.5 *Corporate university and collaborative work*

In the context of CUs, CW is the core that connects internal and external stakeholders (employees, managers, suppliers and partners), creating customer value, improving performance and building competitive advantages (Han et al., 2024). CUs that integrate KM facilitate leadership and external partnerships (Amoozad Mahdiraji et al., 2022), knowledge exchange and innovation (Araujo & Domingos, 2020), and they have been found to strengthen networks, corporate memory and the development of learning cultures (Li & Alagaraja, 2007; Margherita & Secundo, 2011).

The need to improve CW has been highlighted in Latin America and the Caribbean. For example, challenges in collaboration between franchises have been identified in Brazil (Rocha et al., 2024), and difficulties in working with stakeholders have been noted in Chile (Acevedo & Diaz-Molina, 2023). These complexities demonstrate that additional research on CW in these novel contexts is necessary. Nonetheless, in developed economies, CW within CUs has been found to facilitate internal collaboration between L&D and other departments (Cappiello & Pedrini, 2017; Chen et al., 2022). Externally, performance is enhanced through partnerships for knowledge exchange and technology adoption (De Almeida et al., 2021). However, scholarly research has yet to provide a framework for CW in emerging economies.

4. Corporate university model and its interactions

The proposed CU model (see Figure 4 below) integrates the definition of CU presented in the literature review and is designed to assist firms: promote strategies for environmental care and the social welfare of their employees (Alonso-Gonzalez et al., 2018); cultivate a corporate culture that values knowledge (Lissillour & Rodriguez-Escobar, 2023); use ICTs, emerging technologies, such as I4.0 and I5.0; strengthen KM processes, enhance revenue, improve customer retention and support employee productivity (Ilyas, 2017). Figure 4 also aims to improve training and knowledge exchange with stakeholders (Bresolin & De Sá Freire, 2023; Chen et al., 2022, 2023). Additionally, this model aims to consolidate inter- and intra-organizational networks and partnerships (Margherita & Secundo, 2011) to improve competitive advantage (Aziz & George, 2020; Han et al., 2024). Overall, this model seeks to improve the current understanding of CUs, exploring its dynamic connections with corporate strategy and the creation of sustainable competitive advantages (Lissillour et al., 2020).

This model can be applied to drive the development of strategic alliances mediated by KM (Bamel et al., 2021), emphasizing that the firm should integrate all KM activities (Abubakar et al., 2019; Nonaka, Toyama, & Konno, 2000). It also recognizes the potential of CUs for every milieu stakeholder (Han et al., 2024). In addition, CUs facilitates the use and appropriation of emerging technologies such as I4.0 (Heredia et al., 2023; Lista Rossetti et al., 2024). Even though similar approaches already exist in the literature (Chen & Xu, 2022; Han et al., 2024), the proposed CU model represents an attempt to bridge the literature

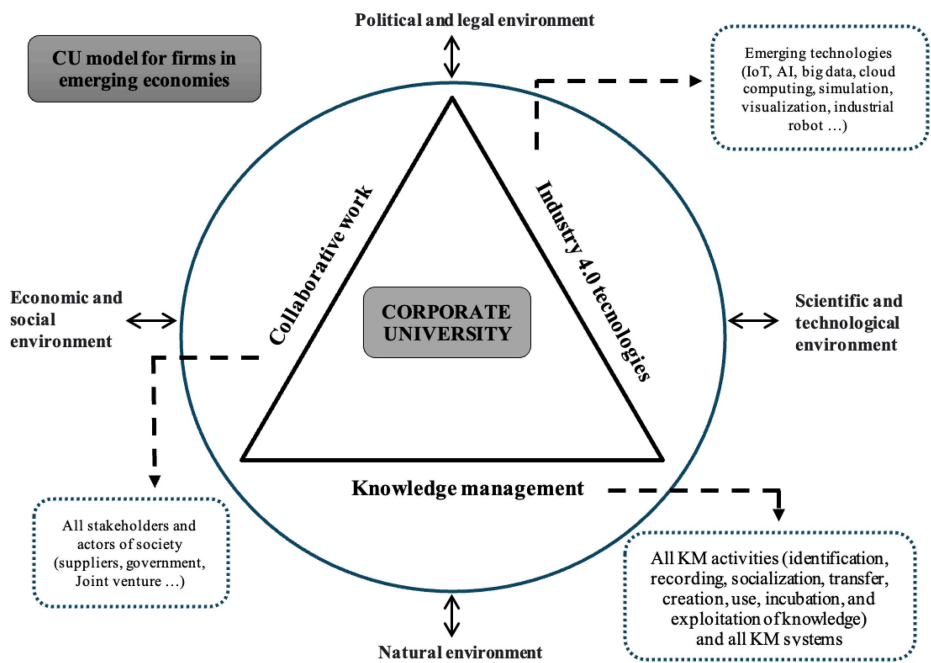


Figure 4. CU model for firms in emerging economies
Source: Authors' own work

in different domains on CU within the same model for emerging economies, providing a holistic and complete view rather than focusing on isolated aspects.

The 2023 CEPAL report, *Social Panorama of Latin America and the Caribbean*, highlights that the proposed CU model addresses the corporate environment in emerging economies. This report outlines structural challenges amid economic, environmental and geopolitical uncertainties. It also notes positive trends, such as reduced poverty, income inequality and increased labor participation. CEPAL (2023) emphasizes the importance of employment in addressing inequality and enhancing social cohesion. The CU model presented in Figure 4 considers these factors and uses KM to help firms identify opportunities, ensure compliance and improve profitability (Heredia et al., 2023). It prepares firms for political, technological and environmental challenges, fostering social responsibility and improving financial outcomes (Heredia et al., 2023; Jamali, 2006). Finally, the model holistically integrates KM, I4.0 and CW, factors often discussed in the literature that focus primarily on developed economies. The next sections outline the relationships of each factor in the CU model.

4.1 Knowledge management impact on firms

Business competitiveness is driven by continuous KM activities (Nonaka et al., 2000). In this stream of the literature, distinctions are drawn between explicit (processed quickly) and tacit (gained through experience) knowledge (Polanyi, 1966). Additionally, KM extends beyond information management, encompassing activities such as knowledge exchange, transfer, absorption, creation (Bamel et al., 2021) and application (Abubakar, Elrehail, Alatailat, & Elçi, 2019). It also involves internalization, externalization (Lui Abel & Li, 2012), SECI and Ba processes (Nonaka et al., 2000) and knowledge elicitation methods, like task analysis and interviews (Hoffman et al., 1995). Consequently, KM is a vital source of competitive advantage for firms, linking data, information and experience (Gibby et al., 2006). It also integrates governance, leadership, knowledge technologies and corporate performance management (Milton, 2007). In recent decades, KM has boosted corporate activity by incorporating digital transformation, open innovation ecosystems, emerging technologies, decision-making, fostering collaboration and efficient KM (De Bem Machado et al., 2022).

In terms of corporate benefits, the literature presents KM as fundamental for corporate survival and growth (Gibby et al., 2006), as it facilitates the identification of opportunities and the efficient use of resources –data, networks, processes and knowledge (Miao et al., 2022); KM also contributes to value creation via the establishment of strategic alliances and partnerships (Bamel et al., 2021). Learning development, for example, the 70:20:10 model improves the experiential, social and structured learning (Jennings, 2012). The integration of internal employees with stakeholders (Asalla et al., 2019; Gibby et al., 2006; Nonaka et al., 2000).

Regarding KM support structures, the “Knoco, 2023” (Knoco, 2023) reveals that various departments, beyond the Chief Knowledge Officer, support KM, including Technology/Innovation/IT. Firms often promote KM through human resources or L&D departments, sharing responsibilities (Jennings, 2012). L&D should enhance efficient learning and strategically align with human resources. In the context of CUs, authors agree that the functions of a CU extend beyond L&D (Lissillour et al., 2020; Rhéaume & Gardoni, 2016). Therefore, CUs should facilitate all KM activities and systems, integrating L&D departments where applicable.

In the proposed model, KM represents a crucial underpinning of effective CUs. Unlike studies that partially examine KM, presenting only some of its activities, this CU model (Figure 4) integrates all KM activities: identification, recording, socialization, transfer, creation, use, incubation and exploitation, including both tacit and explicit knowledge across

internal and external environments (Bamel et al., 2021; Milton, 2007; Nonaka et al., 2000). For example, CUs can enhance KM by applying methodologies to extract tacit knowledge, helping new employees learn from experts through interpersonal exchange (Asher & Popper, 2021), by promoting a knowledge culture within the firm (Asalla et al., 2023). Thus, integrating tacit knowledge into KM systems involves people, technologies and processes (Lui Abel & Li, 2012). According to Nonaka et al. (2000) a comprehensive KM approach is essential for organizations to identify both explicit and tacit knowledge that aligns with their mission and vision. It helps document key lessons learned and create knowledge maps for easy access. Additionally, it supports the socialization and transfer of knowledge, which are critical for building cohesive teams and fostering partnerships. A robust KM strategy also enables the creation of new knowledge, reducing dependence on external sources. Ultimately, it ensures that knowledge is leveraged effectively to enhance the capabilities of personnel.

The KM activities listed above are widely known and applied by firms in developed economies to foster competitive advantage (Chen & Xu, 2022). However, KM is often undervalued in organizations operating in emerging economies, restricting innovation and competitiveness. Raising awareness, particularly among managerial staff, about KM's significance in decision-making and corporate performance is essential. Internal and external knowledge is now recognized as a critical resource for firms and nations to address challenges and promote societal progress (Chen et al., 2019, 2022; Rhéaume & Gardoni, 2016). For CU effectiveness, KM, including all the activities (identification, recording, socialization, transfer, creation, use, incubation and exploitation), must provide the underpinning for CU effectiveness (Asalla et al., 2019; Gölgeci et al., 2019).

4.2 Emerging technologies: Industry 4.0 and Industry 5.0

Originating in Germany, I4.0 focuses on intelligent manufacturing, real-time data and product customization, improving efficiency through technologies such as IoT, big data and cybersecurity (Anshari et al., 2022; Capestro & Kinkel, 2020). Firms adopting I4.0 must enhance personnel capabilities through KM to achieve sustainable competitive advantage (Anshari & Hamdan, 2022; Kwon & Kim, 2020).

Technological developments undoubtedly impact individuals, firms, nations, society and the environment. The proposed CU model incorporates the concept of emerging and advanced technologies, including I4.0 and I5.0. Its use is not limited to KM (Lista Rossetti et al., 2024), but also encompasses the training of personnel, especially those in management functions. In addition, companies can use these technologies in various processes, such as productive, financial, commercial and human talent (Capestro & Kinkel, 2020). Integrating emerging technologies into CUs enables firms to respond to environmental changes, drive innovation, seize business opportunities (Chen et al., 2019) and foster continuous learning, upskilling and adaptability (Pasumarthi et al., 2024).

I4.0 is predominantly implemented in firms from developed economies, particularly in the automotive and service sectors (Demeter et al., 2020). Integrating I4.0 and KM in organizations in Vietnam has been found to enhance competitive advantages, customer loyalty and service offerings while fostering communication and knowledge creation (Duarte Alonso et al., 2024). Additionally, according to Pandya et al. (2023), the use of I4.0 is relevant for business sustainability. However, Shakur et al. (2024) and Metwally et al. (2024) point out that the adoption of I4.0 imposes great challenges in terms of technological investments and value chain infrastructure in emerging economy companies. Although I4.0 adoption has been demonstrated to improve firm performance, requiring skilled employees and practical implementation, research in Latin America and the Caribbean remains limited

(Kamble, Gunasekaran, & Dhone, 2020). Thus, for this model, emphasis is placed on I4.0 and I5.0 (Capestro & Kinkel, 2020; Pasumarthi et al., 2024), technologies that impact people, organizations, nations and society.

4.3 Collaborative work – CW

The current competitive environment, as well as the expansion of ICTs, drives strategic collaboration, challenging firms to develop their abilities to create and use knowledge to strengthen sustainable competitive advantages (Ribeiro & Nagano, 2023). Likewise, through the collaboration that drives CUs, organizations are more likely to form strategic alliances and gain access to new markets and diversify their product portfolio (Bamel et al., 2021), CW also enhances knowledge exchange, while technologies combined with CW, facilitate co-innovation and increase purchasing power and market impact (Niu, 2010). The literature highlights the main advantages of CW, demonstrating that it improves strategies through corporate networks, supports the development of dynamic capabilities essential for successful collaboration (e.g. university–industry–government), enhances KM and positively impacts corporate performance by enabling knowledge exchange (Ribeiro & Nagano, 2023). Similarly, CW has been integrated through KM internally or externally in the context of the CU. In most cases, collaboration is linked internally within the firm (Ayuningtias et al., 2016) and, in fewer cases, externally with some stakeholders (Chen et al., 2022; Margherita & Secundo, 2011; Silva & de Sá Freire, 2017). However, research shows that global challenges encourage companies to collaborate and form strategic alliances with stakeholders and other societal actors, increasing access to new markets, generating value and diversifying products (Bamel et al., 2021).

The development of CW has increasingly gained recognition as a strategy to foster the sustainable competitive advantages of firms, regions and countries in developed economies (Faustino et al., 2019). This is due to creating business alliances and networks, facilitating resource exchanges; reducing the risks associated with innovations, increasing response times and decreasing costs; and allowing adaptation to environmental challenges (Acevedo & Diaz-Molina, 2023; Tortorella, Cawley Vergara, Garza-Reyes, & Sawhney, 2020). In contrast, emerging economies show lower CW due to a lack of trust (Ayuningtias et al., 2016; Heredia et al., 2023), diverging agendas and timelines between businesses and academia. There is also limited alignment among the state, companies and academic actors and constrained internal and external KM capabilities (Aziz & George, 2020; Faustino, Gohr, & Santos, 2019). Finally, in emerging economies, firms must develop KM strategies to survive (Zhao et al., 2022). The growing reliance on complex technologies reshapes workforce organization, requiring new skills and work environments while challenging traditional industrial models (Saniuk et al., 2024). Enterprises recognize the need to adapt to modern technologies and digitalization. The proposed model positions CW as a key solution, helping firms navigate change, drive innovation and collaborate effectively with stakeholders (Chin et al., 2019).

In summary, CW contributes to effective KM, improves organizational performance, fosters growth (Abdelwhab Ali et al., 2019), strengthens relationships with strategic partners (Rhéaume & Gardoni, 2016; Scarso, 2017); and favors the development of human talent, as well as the optimization and integration of resources (Capestro & Kinkel, 2020).

5. Discussion

The CU model proposed in this study highlights the importance of effective KM (explicit and tacit, internal and external, etc.) (Asher & Popper, 2021) in a globalized, competitive, dynamic and uncertain environment. It also demonstrates the need to incorporate emerging technologies (I4.0 and I5.0), such as robotics, artificial intelligence, virtual and augmented reality and big

data, as well as collaboration with different stakeholders (investors, collaborators, customers, suppliers and other actors in society such as companies, academic institutions and government entities), to improve competitive advantage. In addition, it emphasizes the importance of considering KM, emerging technologies and CW holistically rather than treating them separately, as has been traditionally evidenced in the literature.

In particular, the model set forth in this study seeks to contribute to improving the sustainable competitive advantages of firms in emerging economies, which are generally characterized by unskilled labor, limited KM structures and ICT integration. Moreover, these organizations tend to prioritize a short-term vision, low-risk tolerance and individual work. These traits limit their competitive capacity (Alonso-Gonzalez et al., 2018).

In the specific case of KM, the literature has demonstrated that KM practices are essential for developing human potential, analyzing the environment, incorporating technologies, fostering collaboration and supporting business decision-making, all of which strongly impact competitive capacity. Therefore, its use is widespread in companies in developed economies (Chen et al., 2022). Furthermore, the literature also shows that, in most of the CU models, management focuses on some KM activities; however, these are often considered in isolation, focusing on all KM activities holistically is likely to yield more positive returns (Abubakar et al., 2019; Chen & Xu, 2022; Nonaka et al., 2000). For this reason, the CU model proposed in this paper emphasizes the importance of a comprehensive approach to KM (Choi et al., 2020), considering the particularities of firms in emerging economies, especially in Latin America and the Caribbean, which are the focus of this research.

Regarding the role of emerging technologies (I4.0 and I5.0) in CU models, the literature reveals that, in most of the models reviewed, emphasis is attributed to ICTs. However, in companies in developed economies, emerging technologies such as artificial intelligence, robotics, virtual and augmented reality, big data and the IoT are being widely implemented due to their great impact on competitive capabilities.

The emphasis on ICTs in CU models is mainly because universities still focus on human talent and administrative functions, with limited impact on other processes such as production and finance, where they could play an important role. Therefore, the proposed model emphasizes the incorporation of emerging technologies (I4.0 and I5.0, ICT) as a strategy, supported by KM and CW, to enhance business competitiveness. In the case of companies in emerging economies, CUs should guide managers in the design of strategies that enable the adoption and use of these technologies to respond competitively and sustainably to the challenges of the environment and identify business opportunities (Capestro & Kinkel, 2020; Chen et al., 2019).

Likewise, the literature evidenced that, in the current business environment, CW allows companies to share experiences and resources, reduce risks, incorporate technologies and increase innovation, all in favor of their competitive capacity (Bamel et al., 2021). For this reason, the proposed CU model incorporates CW as one of the key variables. This is especially relevant in companies in emerging economies, where resources are limited, risks are high, innovation is almost nil, technology incorporation is very limited and individuality predominates. This type of work, supported by KM and technologies, is of great benefit to these companies, since the change from an individual to a collective organizational culture will bring numerous benefits. CU has a very important role in this process (Han et al., 2024).

6. Concluding remarks

The SLR conducted for this study shows that CUs are a relevant business strategy, increasingly recognized by organizations in developed economies for enhancing competitiveness. In today's

era, marked by rapid technological advancements, unpredictable and accelerated changes across diverse environmental dimensions and rising sustainability demands, companies require innovative strategies and CUs are among them. In contrast, among companies in emerging economies, the concept and appropriation of CUs are poorly understood by academics, executives and decision-makers, leading to missed opportunities to leverage their benefits. Accordingly, the CU model proposed here is specifically designed for companies in emerging economies, typically characterized by limited appreciation of knowledge (internal and external), centralized decision-making, low risk tolerance, scarce technological investment and a strong individual work culture. In summary, this model emphasizes effective KM, strategic technological adoption and CW as essential variables for building competitive advantage, regardless of sector or environment, addressing the particular challenges faced by companies in emerging contexts.

6.1 *Contributions to theory*

This paper contributes to the CU literature in different ways. First, the model developed in this study seeks to promote reflection and to advance the literature on CUs in emerging economies. In these economies, the KM implemented generally comes from developed countries and requires significant adjustments that are often not made. Owing to this lack of adaptation, the results are often not as expected compared to those obtained by companies in developed economies. Second, it expands the limited research on CU models in emerging economies by integrating KM, I4.0 and CW. It illustrates how CUs incorporate KM activities and systems, it has presented a holistic definition of CU. Third, while existing models emphasize technology use, this study highlights the importance of emerging and advanced technologies, such as I4.0 and I5.0, which enhance firms' sustainable competitive advantages and domain knowledge. Fourth, it broadens the debate on CUs beyond developed economies, confirming the need for internal and external strategies in emerging economies to address current changes and challenges agilely. The CU model promotes CW through alliances, networks and partnerships, enabling resource sharing, improved innovation capacity, faster responses, reduced costs and better environmental adaptability.

6.2 *Contributions to practice*

The model presented in this paper shows to executives and corporate decision-makers the benefits of having a CU within their companies, understood as a strategy that enables organizations to enhance their competitive capacity in a sustainable way through effective KM, the appropriate use and adoption of emerging technologies (such as artificial intelligence, robotics, virtual and augmented reality, 3D printing, the IoT, data analytics) and CW with various internal and external stakeholders.

Likewise, the model shows that the CU is a strategy to support the full range of business activities – including human talent development, optimization of productive or service processes and commercial and financial effectiveness. All of this is aimed at positively impacting the social, environmental and economic dimensions of the company and its surrounding environment. Finally, the model highlights to consultants and academics interested in supporting companies in strengthening their competitive advantage the benefits that the creation of a CU brings for that purpose and for its stakeholders, as well as the key aspects to consider in its implementation.

6.3 *Limitations and recommendations for further research*

The limitations of this study are: first, the literature review was limited to two databases; expanding the search to more databases and considering literature in various languages

would be beneficial. Second, according to the literature on CUs in the particular context of emerging economies, the proposed model presents only three relevant factors, which may disregard additional complexities and factors related to CUs. Future research should explore other variables pertinent to CUs in these contexts. Third, in emerging economies where KM is not a priority, this model faces the limitation of low adoption; however, it could persuade management to integrate KM more significantly. Finally, as CUs represent a constantly changing phenomenon, researchers should be careful when generalizing the results of this research to other CU settings.

Given the gaps prevalent in the literature, future scholars should consider conducting more empirical evidence on CU variations. Future studies should validate the CU model through comparative case studies analyzing successful implementations, explore the impact of emerging technologies and assess long-term outcomes in the context of emerging economies. In addition, the proposed model suits large organizations, as their structure enables better decision-making through CU-driven KM. Smaller firms can also apply the model; however, they should do so progressively. Additionally, it is important to consider that in these organizations, it is necessary to have an advisory group in charge of coordinating the integration of KM in the whole organization.

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Supplementary material

The supplementary material for this article can be found online.

Appendix

Table A1. CU models factors present

Author	Factors present in the model			
	KM	Technology	I4.0	CW
Han et al. (2024)	Yes	No	No	Yes
Chen et al. (2023)	Yes	No	No	No
Bresolin & De Sá Freire (2023)	Yes	Yes	No	Yes
Chen, Xu, & Zhai (2022)	Yes	No	No	Yes
Chen & Xu (2022)	Yes	Yes	No	Yes
Lissillour, Rodríguez-Escobar, & Wang (2020)	Yes	Yes	No	Yes
Araujo & Domingos (2020)	Yes	Yes	No	Yes
Alonso-Gonzalez et al. (2018)	Yes	Yes	No	No
Ilyas (2017)	Yes	No	No	No
Silva & de Sá Freire (2017)	Yes	Yes	No	Yes
de Sá Freire et al. (2016b)	Yes	Yes	No	Yes
Ayuningtias, Nurbaiti, Anggadwita, & Kriseka Putri (2016)	Yes	Yes	No	Yes
Vives (2015)	No	Yes	No	Yes
Kolo, Strack, Cavat, Torres, & Bhalla (2013)	No	Yes	No	Yes
Margherita & Secundo (2011)	Yes	Yes	No	Yes
Li & Alagaraja (2007)	Yes	Yes	No	No
El-Tannir (2002)	No	Yes	No	No
Prince & Stewart (2002)	Yes	Yes	No	No
Beaver & Prince (2001)	Yes	Yes	No	No

Note(s): Supplementary Files 1–6 are given in a separate document, please check
Source(s): Authors’ own work

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